

Day 13 HW: Soccer 2v2 Geogebra Questions

Directions: Use the applet on the provided Geogebra website:

<https://www.geogebra.org/m/zrydzrsu#material/cpgehp7y>

Follow the instructions of each problem scenario and provide your answers to the following questions below.

1. OffenseA has the ball, and needs to make a split-second decision. She is deciding between shooting or passing to her teammate, OffenseB.

Which do you think is the better decision and why? Explain your reasoning in 2-3 sentences.

assuming that the defense wont move, it would make sense to just shoot. The reason for this is that the defender on her teammate is closer to the teammate then the goalkeeper is for her so she does not need to be as accurate to shoot.

2. Part 2: Investigate

- a. She takes the shot:

Suppose OffenseA has the ball and **decides to shoot**.

How much of the goal is covered by the goalkeeper's dilation?

Maybe a bit less then half

Where does the dilation of DefenderA end up if we use this point as the center of our dilation?

it is completely off the field, not doing anything

- b. She passes:

Change the checkboxes so that OffenseB has the ball now.

How did this change the images on the dilation?

Now, rather than have about half the goal, they have only a fraction on each side

Are they in the same spot as in 2a?

No, they are where offense b is not where offense A is

Describe how the images are positioned compared to the center of dilation.

the closer the center of dilation the harder the shot

c. Other observations:

What other geometric properties (**angles, area, midpoints, parallel lines**) did you find in the scenario? Record your observations.

perpendicular line from offense a to goal when offense a has ball

3. Part 3: Revise

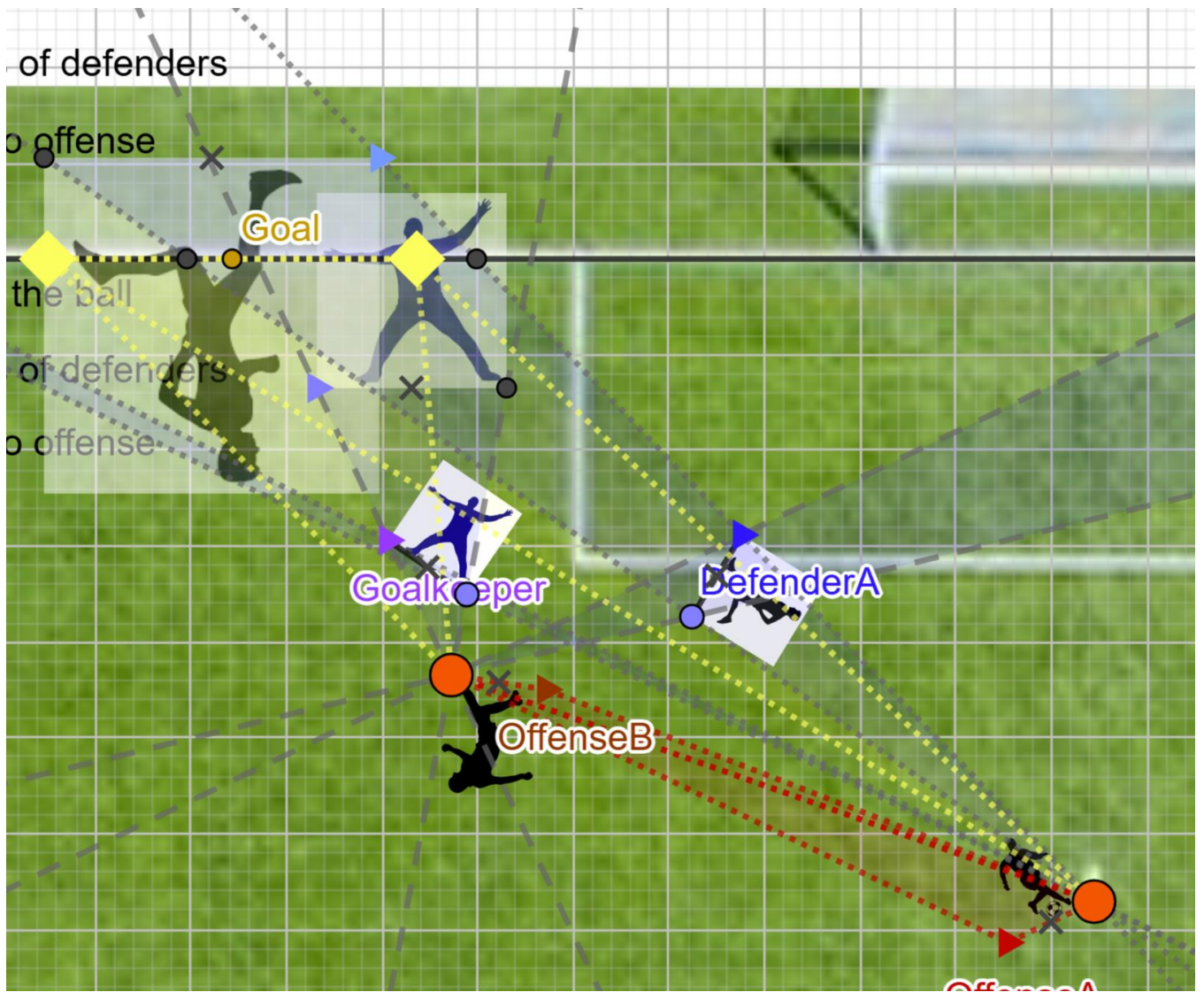
Now revisit your initial answer.

Would you change your advice from part 1, or make the same suggestion? **Justify** your response with some of your observations from part 2.

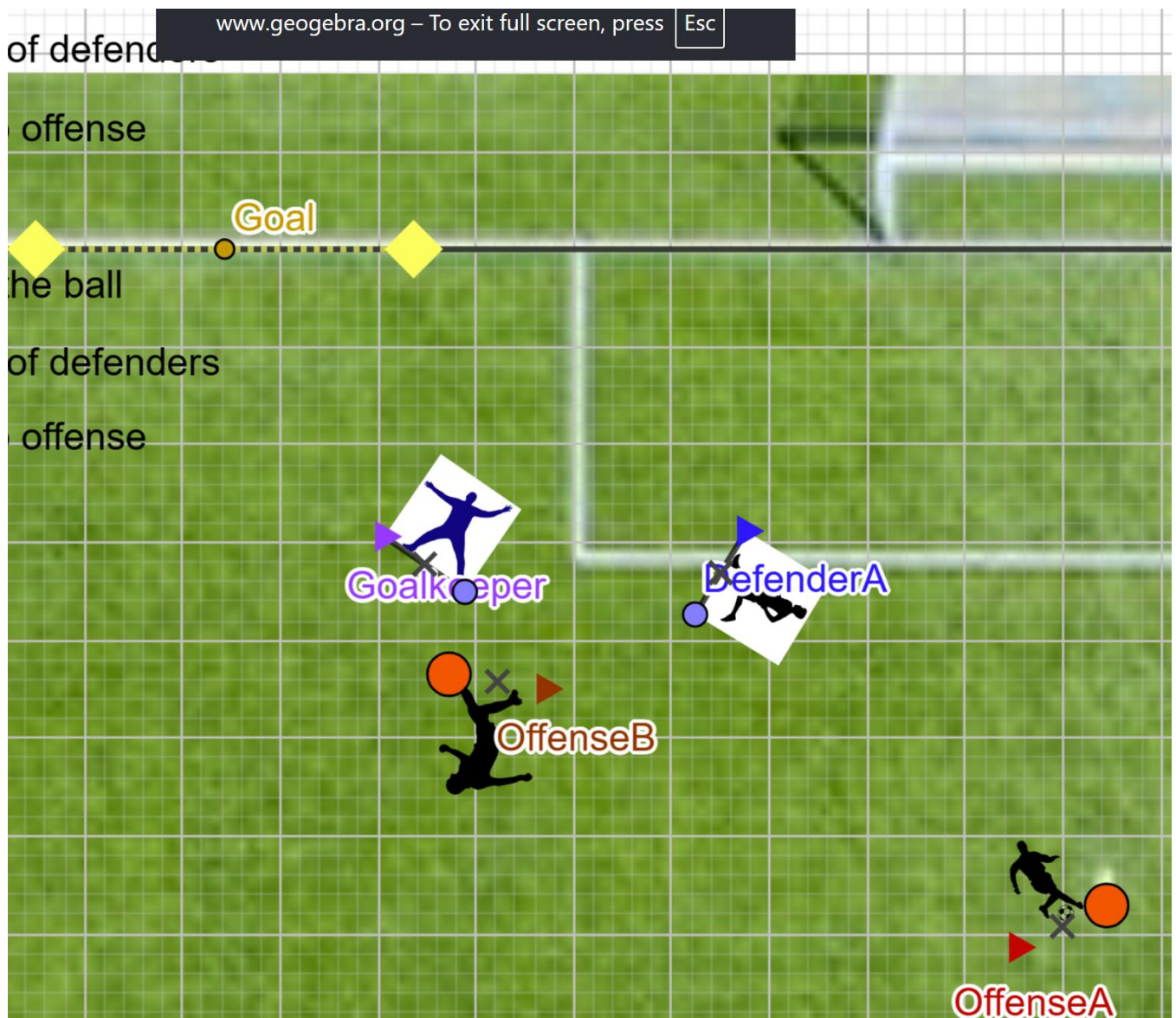
part two gave me more info on what I eyeballed and now I know for sure it would be much easier for one to shoot if they don't move, like I said earlier

4. Part 4: Create

Design a different "Pass or shoot" scenario in the applet and provide a snippet in the space below. Explain who has the ball, which choice you'd recommend (pass or shoot), and why.



5.



6.

Offense A has the ball, and the play is to pass to Offense B because Offense B has a slightly better angle and more possible shots